

DATA STRUCTURES AND ALGORITHMS

1. DSA / Placement Related Tasks

- Find out the list of companies which are conducting placement drives using DSA course.
- Find out the syllabus and gate for DSA.
- Find out the set of inputs and outputs for the following:
 1. Electricity Bill
 2. Bus Ticket
 3. Message to a Friend
- Today, when you started from home to college/hostel, what are the various data structures you identified in your surroundings?

2. GATE Syllabus – Programming and Data Structures

1. Programming & Data Structures

- C programming basics
- Recursion
- Arrays
- Stacks
- Queue
- Linked Lists
- Trees
- Binary Search Trees
- Binary Heaps
- Graphs

2. Algorithms

- Searching – Linear Search, Binary Search
- Sorting – Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Heap Sort

- Hashing
- Time and Space Complexity – Big-O (O), Omega (Ω), Theta (Θ)
- Algorithm Design Techniques – Divide and Conquer, Greedy Algorithms, Dynamic Programming
- Graph Algorithms – BFS, DFS, Minimum Spanning Tree, Shortest Path

3. Companies for DSA Preparation / Placements

Tier 1 (Very High DSA):

- Google
- Microsoft
- Amazon
- Meta
- Apple
- NVIDIA
- Uber
- Atlassian
- Adobe
- Salesforce

Tier 2 (Strong DSA):

- Flipkart
- Walmart
- Intuit
- Oracle
- PayPal
- Visa
- American Express
- Goldman Sachs
- Morgan Stanley

- D.E. Shaw

Indian Product Companies:

- PhonePe
- Razorpay
- Meesho
- Groww
- CRED
- Freshworks
- Zoho
- Juspay
- Swiggy
- Zomato

Service-Based Companies (Basic to Moderate DSA):

- TCS
- Infosys
- Wipro
- Cognizant
- Accenture
- Capgemini
- HCLTech
- Tech Mahindra
- IBM

4. Set of Inputs and Outputs

A. Electricity Bill

Input:

- Consumer Number
- Customer Name
- Previous Meter Reading
- Current Meter Reading
- Rate per Unit
- Fixed Charge

Output:

- Units Consumed
- Electricity Bill Amount
- Due Date
- Payment Status

B. Bus Ticket

Input:

- Passenger Name
- Source
- Destination
- Date
- Bus Type
- Number of Tickets

Output:

- Ticket Number
- Passenger Name
- Seat Number
- Fare

- Bus Number
- Journey Date

C. Sending a Message to a Friend

Input:

- Sender Name
- Receiver Name
- Message
- Date
- Time

Output:

- Receiver
- Message
- Date & Time
- Status

5. Data Structures Identified in Surroundings

- Mess token – Object
- Plates – Array
- Classroom benches – Array
- Books – Array
- Subjects – Array
- Attendance register – Array